

SatClip: ask a question about land, get an answer you can check.

Free satellites photograph all of India every few days. SatClip lets anyone ask a plain question about a place and a date, and answers with a measurement taken from those images, the proof behind it, and an honest "not enough evidence" when it cannot be sure.

The problem

Satellite images could answer urgent questions like "which villages are flooded?" or "did this crop fail?". Today, using them is hard:

- **It needs specialist skills.** Choosing satellites, downloading huge files and processing them takes mapping (GIS) training most offices do not have.
- **Clouds hide the ground** in the monsoon, exactly when floods happen. Ordinary photos from space show only cloud.
- **Official maps are shared thin.** In 2024 India's national flood-mapping service issued about 300 flood maps for the whole country, so most local questions go unanswered.
- **AI chatbots sound sure but show no proof.** A fluent answer with no date, image or confidence is hard to act on.

Who it helps

District disaster officials "Which parts of my district are under water since Tuesday, and is it spreading?"

Agriculture and crop-insurance officers "Did crops in these blocks really suffer this season?"

Journalists and fact-checkers "Was this place really flooded on the date a viral post claims?"

Planners and researchers "How much has this town grown since 2020?"

We start with **monsoon floods and crop condition at district level**, where a slow or wrong answer costs the most.

How it works, in four steps

1

You ask

Type a question in plain words. SatClip repeats back what it understood, so a misreading is caught at once.

2

It finds the right image

It searches free public satellite archives. If clouds block the view, it switches to radar, which sees through cloud.

3

It measures

A transparent, well-tested method does the measuring (for example, spotting water in radar images). The AI never makes up the number.

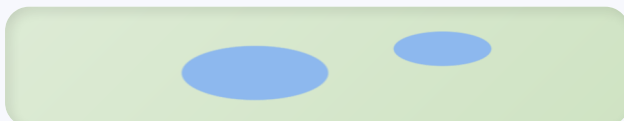
4

It shows the proof

You get an evidence card: the answer, a map, which image and date it came from, and how confident it is.

WHAT AN ANSWER LOOKS LIKE

"How much of Barpeta district was under water around 11 July 2024?"



About ___ sq km under water

Confidence: high

Radar image, 11 Jul 2024

Scene ID shown

Re-run receipt

Illustration of the layout only. The measuring engine is being built now, so no real figure is shown.

And when it is not sure

If the image is unclear, the question is outside what SatClip can measure, or confidence falls below a set level, it says "**insufficient evidence**" instead of guessing, and tells you what would help, such as the date of the next satellite pass.

A careful "I don't know" protects decisions. A confident wrong answer about a flood or a crop can send help to the wrong place or deny a fair claim.

Why it is easier than what exists

Our innovation is not a bigger AI model. It is a shorter, checkable path from a question to an answer someone can act on.

	TYPICAL SATELLITE AI CHATBOT OR GIS TOOL	SATCLIP
Skills needed	GIS training, or knowing which satellite and settings to use	Just a question in plain words
Monsoon clouds	Often no usable image, or you must know to switch to radar	Switches to radar automatically and says so
Where the number comes from	Generated by the AI model itself	Measured by a transparent method from the actual image
Proof	Usually none shown	Image ID, date, map, confidence and a receipt anyone can re-run
When unsure	Usually answers anyway	Says "insufficient evidence" and what would fix it
Cost and setup	Paid licences, powerful hardware, or region-locked services	Free public data, open models, runs on an ordinary computer, can run offline

Why you can trust it

- **Every answer points to its source:** the exact satellite image, its date and the area measured.
- **Confidence is checked, not guessed.** It is tuned against cases where the true answer is known, and shown in plain words: high, moderate or low.
- **Anyone can re-run it.** Each answer has a receipt that reproduces the same result.
- **Evidence first, chat second.** Studies show AI explanations can make people accept wrong answers as readily as right ones, while a well-tuned confidence helps them judge when to rely on it. So SatClip leads with the map and proof.

How it grows

- **More users:** big areas are split into small tiles worked on in parallel; repeated questions reuse results.
- **More places:** it reads only the part of each image it needs, with backup archives if one is down.
- **More languages:** only the question reading changes; numbers never pass through translation.
- **Government use:** runs on a laptop, a state data centre, or fully offline.

What it will not do

Count cars or houses (the free images are too coarse), predict the future, or replace official maps and field checks. It answers the local follow-up questions those cannot.

Where the project is today

Done

 Research review: 75 papers so far, of about 200 planned

Done

 System design and working app skeleton

Done

 Connection to free satellite archives, with automatic radar fallback

Next

 Measuring engine: floods, crop change, land cover

Then

 Full app screens, training code, testing, demo

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Words used here

Radar (Sentinel-1): a European satellite that sends its own signal, so it sees through cloud and at night.

Optical (Sentinel-2): a satellite camera, like a photo from space; blocked by cloud.

GIS: mapping software that usually needs training.

Confidence: how likely the answer is to be right, checked against known cases.